

Average speed

Total distance over total time — and why it is not the average of the two speeds.

LESSON 126–128

3 × 40 MIN

MATEMATIKA 6, §27

S7 12 RATES

LESSON CLOCK

25'

30'

35'

25'

5

The definition	25 min
Journeys in stages	30 min
Why not the mean of the speeds	35 min
Stops	25 min
Homework	5 min

LEARNING OBJECTIVES

- Define average speed as total distance over total time.
- Compute it for a journey in stages.
- Explain why the mean of the speeds is usually wrong.
- Include stops in the total time when the question requires it.

TEXTBOOK

National	pp. 369–378
Cambridge	Stage 7, pp. 122–126
Workbook	Exercise 12.8

01

Explanation

The definition

average speed = $\frac{\text{total distance}}{\text{total time}}$

JOURNEY	TOTAL DISTANCE	TOTAL TIME	AVERAGE SPEED
2 h at 60, 3 h at 80	360 km	5 h	72 km/h
120 km at 60, 120 km at 40	240 km	5 h	48 km/h
90 km at 45, 60 km at 60	150 km	3 h	50 km/h

ONE DIVISION, AT THE VERY END

Find the whole distance, find the whole time, divide once. Every average-speed problem is that sentence.

Journeys in stages

STAGE	SPEED	TIME	DISTANCE
first	45 km/h	2 h	90 km
second	60 km/h	1 h	60 km
total	50 km/h	3 h	150 km

THE AVERAGE LIES BETWEEN THE TWO SPEEDS

Here between 45 and 60, and nearer the one that took longer. If your answer falls outside the two speeds, something has gone wrong.

Why not the mean of the speeds

A car goes 120 km at 60 km/h and comes back at 40 km/h. The mean of the speeds is 50; the average speed is not.

STAGE	DISTANCE	SPEED	TIME
out	120	60	2 h
back	120	40	3 h
total	240	—	5 h

$$\text{average} = \frac{240}{5} = 48 \text{ km/h, not } 50$$

MORE TIME IS SPENT AT THE SLOWER SPEED

That is why the average is pulled below the mean. Only when the two *times* are equal does the mean of the speeds happen to be right.

Stops

QUESTION	TIME USED	ANSWER
150 km driven in 2 h with a 30 min stop: average driving speed	2 h	75 km/h
the same journey: average speed for the whole trip	2.5 h	60 km/h

READ WHETHER THE STOP IS INCLUDED

“Average speed for the journey” includes the stop; “average driving speed” does not. The same numbers give 60 or 75 depending on which was asked.

02

Worked examples

EXAMPLE 1 A car drives 2 h at 60 km/h and 3 h at 80. Find the average speed.

① Distance: $120 + 240 = 360$ km.

② Time: 5 h.

③ $360 \div 5 = 72$ km/h.

Between 60 and 80 ✓

ANSWER

72 km/h

EXAMPLE 2 A car goes 120 km at 60 km/h and returns at 40. Find the average speed for the whole trip.

① Out: 2 h. Back: 3 h.

② Total: 240 km in 5 h.

③ 48 km/h.

Not 50 — more time at the slower speed.

ANSWER

48 km/h

EXAMPLE 3 A bus covers 150 km in 2 hours of driving, with a 30-minute stop. Find both averages.

① Driving: $150 \div 2 = 75$ km/h.

② Whole trip: 2.5 h.

③ $150 \div 2.5 = 60$ km/h.

ANSWER

75 and 60 km/h

03 Interactive model

Ask for the average of 60 and 40 first; almost everyone says 50, and the table that follows is then genuinely surprising.

Total over total

Average speed is:

the mean of the speeds

total distance over total time

the largest speed

the smallest

2 h at 60 and 3 h at 80 gives:

70

72

140

68

120 km at 60 and back at 40 gives:

50

48

52

100

The answer is below 50 because:

the distances differ

more time is spent slower

of rounding

of the stop

90 km at 45 and 60 km at 60 gives:

50

52.5

55

105

An average speed must lie:

above both speeds

between them

below both

anywhere

150 km in 2 h driving plus a 30 min stop, whole trip:

75

60

50

120

The mean of the speeds is right only when:

the distances are equal

the times are equal

always

never

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Average speed	O'rtacha tezlik	Средняя скорость
Total distance	Umumiy masofa	Общее расстояние
Total time	Umumiy vaqt	Общее время
Stop	To'xtash	Остановка
Stage	Bosqich	Участок
Mean of speeds	Tezliklarning o'rtachasi	Среднее скоростей
Weighted	Vaznli	Взвешенный
Journey	Sayohat	Поездка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Average speed uses:

the mean of the speeds

total distance and total time

the first speed

the last speed

240 km in 5 h is:

48

50

1200

4.8

The average of a journey out at 60 and back at 40 is:

exactly 50

less than 50

more than 50

100

An answer of 90 for stages at 45 and 60 is:

possible

impossible

exact

the mean

“Average driving speed” excludes:

the distance

the stops

the return

nothing

Which weights the average?

the distances

the times

the speeds

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 360 km in 5 h

ANSWER 72 km/h

2. 240 km in 5 h

ANSWER 48 km/h

3. 150 km in 3 h

ANSWER 50 km/h

4. 150 km in 2 h

ANSWER 75 km/h

5. 150 km in 2.5 h

ANSWER 60 km/h

6. Average speed is total distance over

ANSWER total time

7. An average must lie

ANSWER between the two speeds

Medium · the standard question

7 PROBLEMS

1. 2 h at 60 and 3 h at 80

ANSWER 72 km/h

2. 120 km at 60 and back at 40

ANSWER 48 km/h

3. 90 km at 45 and 60 km at 60

ANSWER 50 km/h

4. 1.5 h at 50 and 0.5 h at 70

ANSWER 55 km/h

5. Why is the return trip average below 50?

ANSWER More time is spent at the slower speed

6. With a 30 min stop in a 2 h drive of 150 km

ANSWER 60 km/h

7. Without the stop

ANSWER 75 km/h

Hard · stretch and reason

7 PROBLEMS

1. Out at 80 and back at 20 over 80 km each way

ANSWER 32 km/h

2. Out at 30 and back at 30

ANSWER 30 km/h

3. 100 km at 50 then 100 km at 100

ANSWER 66.7 km/h

4. A journey of 3 equal parts at 30, 60 and 60

ANSWER 45 km/h

5. When does the mean of two speeds give the right answer?

ANSWER When the two times are equal

6. A walk of 2 km at 4 km/h and 2 km at 6

ANSWER 4.8 km/h

7. Is the average speed ever above both speeds?

ANSWER No

07 Homework

Homework — 5 tasks

Find the total distance and the total time before dividing once.

1. A car drives 3 h at 50 km/h and 2 h at 75. Find the average speed.
2. A journey of 60 km out at 30 km/h and back at 60. Find the average speed.
3. Explain in one sentence why the answer to task 2 is not 45.
4. A bus covers 180 km in 3 hours of driving with a 45-minute stop. Find both averages.
5. A walk of 3 km at 6 km/h and 3 km at 3. Find the average speed.